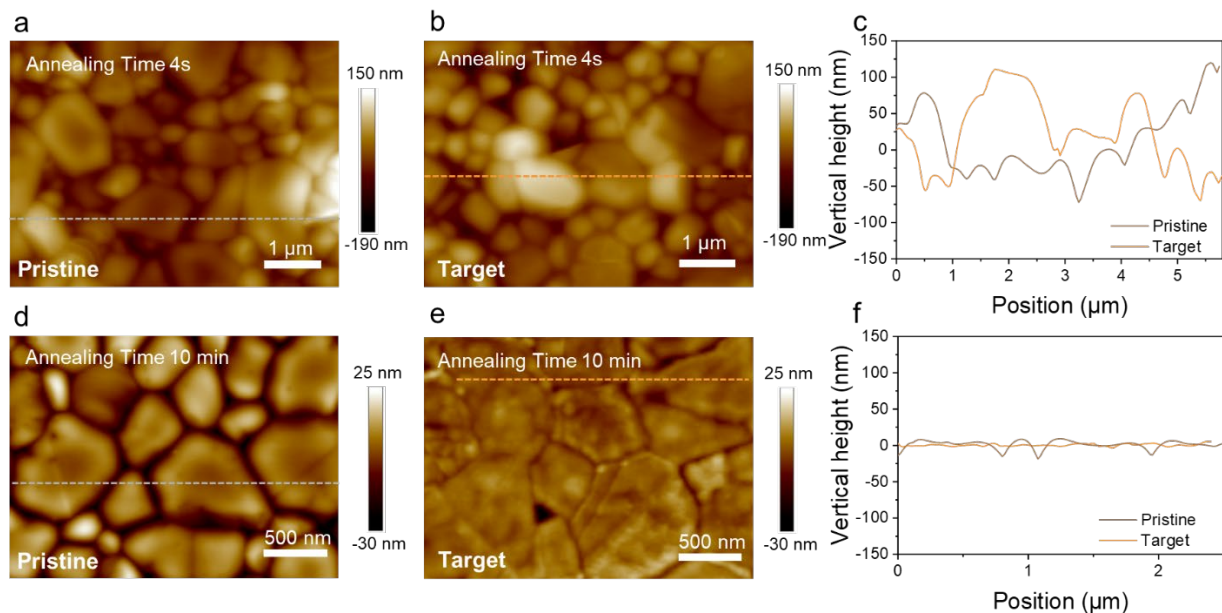
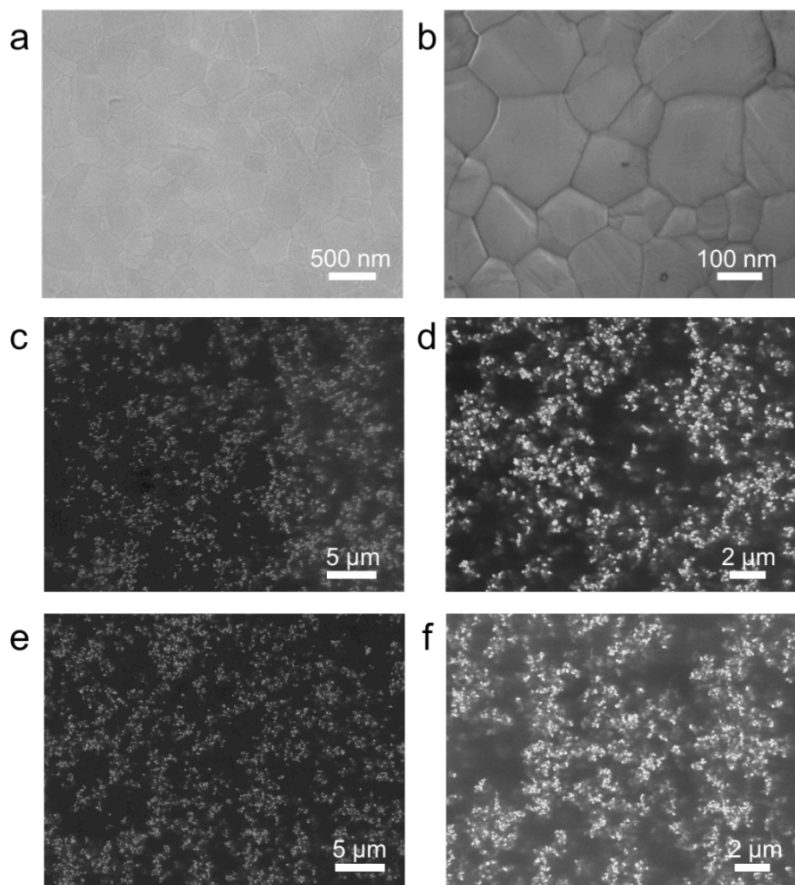

Nanoscopic cross-grain cation homogenization in perovskite solar cells

In the format provided by the
authors and unedited

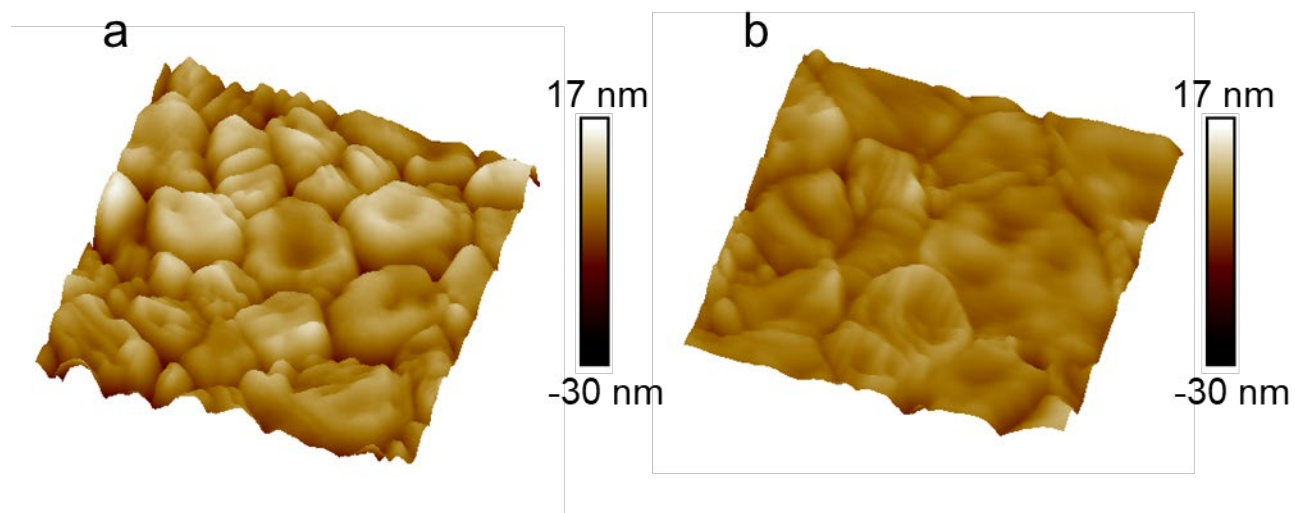
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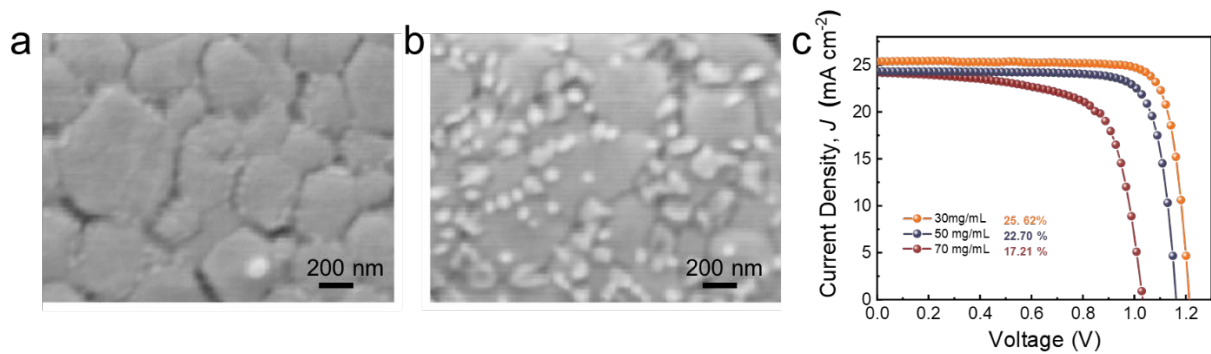
Supplementary Fig. 1 Morphological evolution of the film bottom surface upon annealing. **a-c** AFM topographic images **a-b** and surface height profiles (**c**) of the pristine and target perovskite film bottoms at the initial stage of annealing (4 s, 150 °C); **d-f** AFM topographic images **d-e** and surface height profiles (**f**) of the pristine and target perovskite film bottoms at the end of annealing (10 min, 150 °C). Below, we elaborate on these results in detail. **a-b** show the film bottom surface topography at the beginning of annealing (4 s, 150°C; when the film has just turned to the black perovskite phase). At this stage, the target and pristine film bottoms exhibit very similar grain morphologies. Also, there are relatively large grain height undulations of up to 200 nm in both films **c**, suggesting that grain growth has not yet been fully constrained by the ETL surface. Once the annealing time reaches 10 min, the grain morphology is stabilized for both films with a typical nano-GT geometry **d-e**. In the target case, the nano-GT is significantly shallowed. Also, **f** shows that the heights of individual grains have become similar at 10 min annealing in both films, reflecting that the grains have fully grown and have now been constrained by the ETL surface. These results confirm that the additive begins to be in effect along with the progression of annealing as we discussed in the main text.



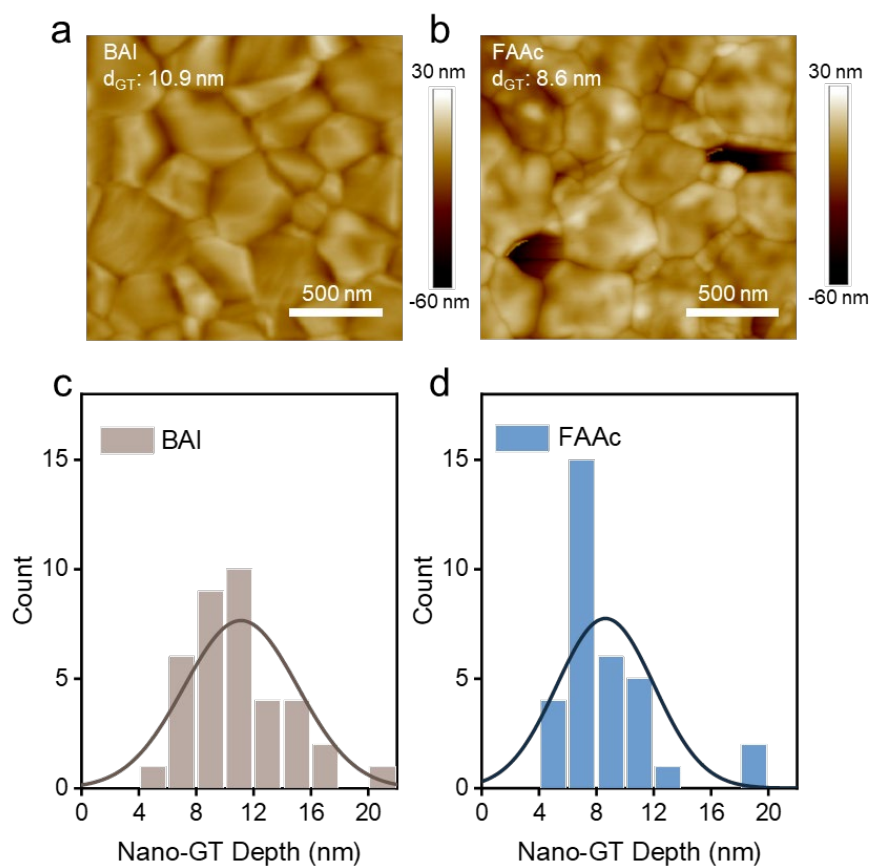
Supplementary Fig. 2 **a-b** SEM images of the pristine **a** and target **b** perovskite film bottoms taken at magnifications of 50,000 $\times$ and 150,000 $\times$; **c-d** SEM images of SnO₂ ETL surface before pristine (**c**) and target **d** perovskite film deposition; **e-f** SEM image of the SnO₂ ETL surface on the delaminated area of the pristine **e** and target **f** film. All the SEM images reveal no occurrence of noticeable trans-crystal fracture during the peeling process. Morphological changes are thus expected to be minimal as long as there is no occurrence of the trans-crystal fracture.



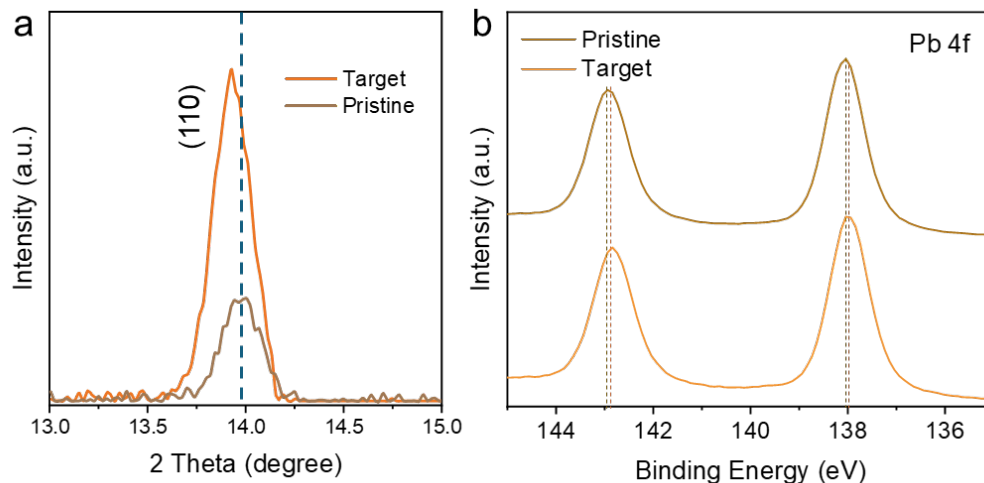
Supplementary Fig. 3 a-b 3D-view AFM images of the pristine **a** and target **b** perovskite films ($1.3\ \mu\text{m} \times 1.3\ \mu\text{m}$). The results show that the nano-GTs in the target film are significantly shallowed across the entire film.



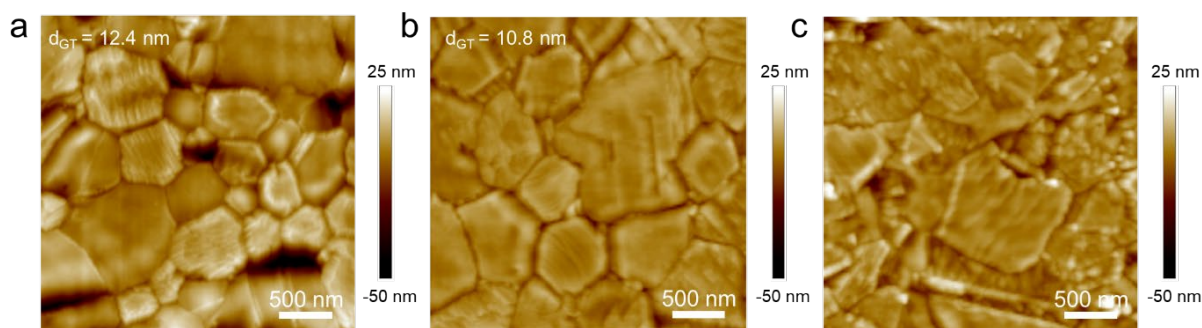
Supplementary Fig. 4 Effects of the BAAC adding amount on the perovskite film. **a-b** SEM images of the perovskite bottom surface after the film delamination from the SnO₂ ETL treated using 50 mg/mL **(a)** and 70 mg/mL **(b)** BAAC in isopropanol, respectively. **c** J - V curves of perovskite solar cells with SnO₂ ETL treated with varying concentrations of BAAC (30 mg/mL, 50 mg/mL, and 70 mg/mL).



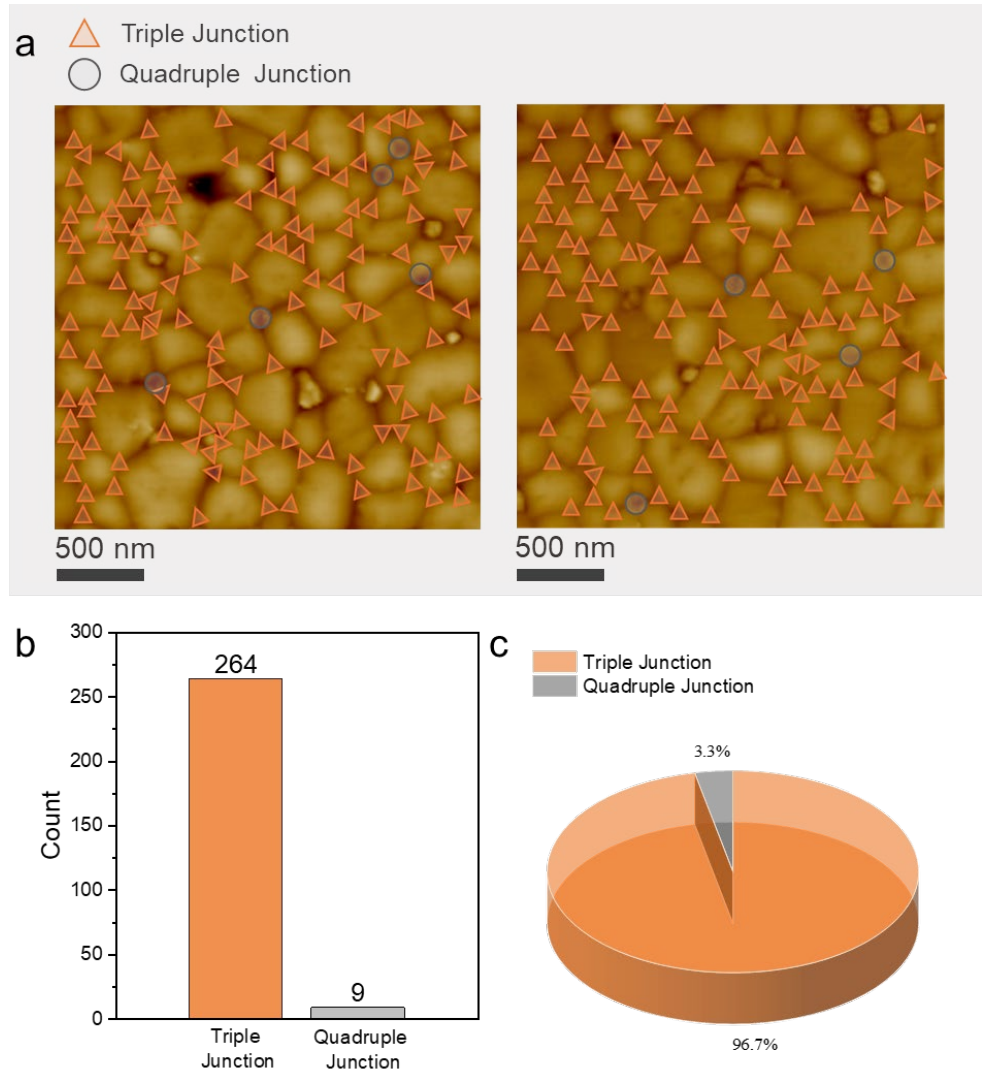
Supplementary Fig. 5 Effects of the additive type on the perovskite film structure. a-b AFM images of the bottom perovskite films made with BAI (a) and FFAc (b) treatment, respectively, and **c-d** the corresponding nano-GT depth statistics.



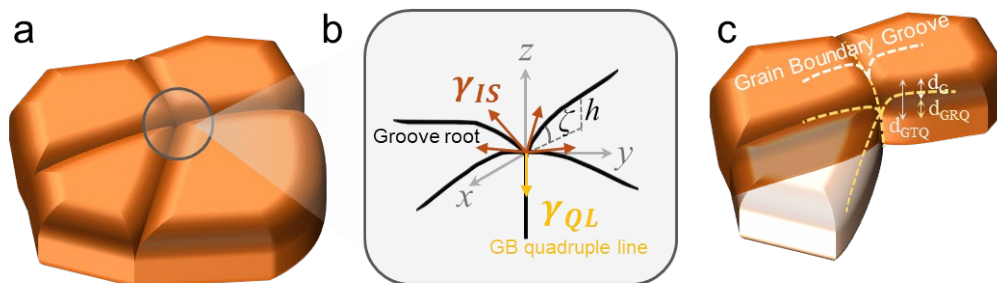
Supplementary Fig. 6 **a** The (110) diffraction peak of the flipped film bottom of pristine and target perovskite films shows a shift of the characteristic XRD peak of perovskite to a lower angle in the target film, attesting to the existence of compressive strain. **b** The XPS spectra of Pb 4f for the flipped film bottom of pristine and target perovskite film. The Pb 4f peak shifts toward a lower binding energy, which can be attributed to the interaction between C=O in acetate and uncoordinated Pb²⁺, thereby reducing the dangling bonds on the surface of perovskite grains.



Supplementary Fig. 7 Effects of BAAC addition in the perovskite precursor solution on the film structure. a-c AFM images showing the bottom surface morphologies of the perovskite films made in the conditions of **a** 2 mol% **b** 5 mol%, and **(c)** 7 mol % BAAC in the perovskite precursor solutions.

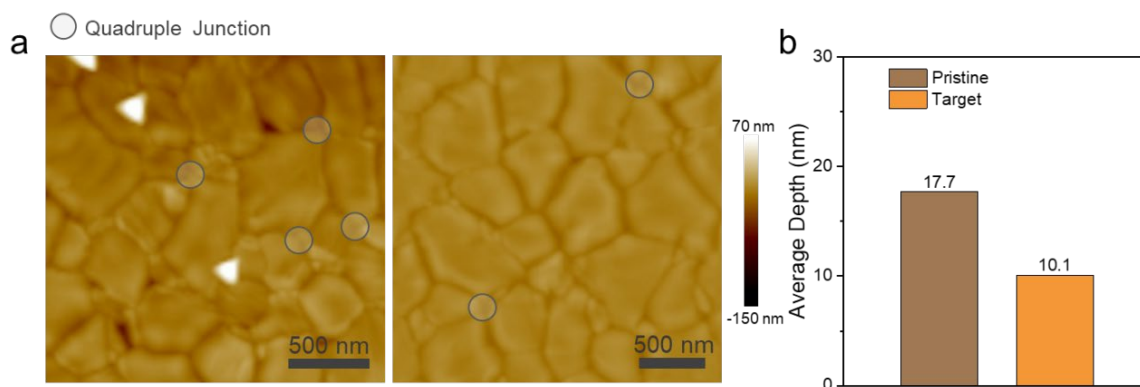


Supplementary Fig. 8 **a** Labelling of triple and quadruple junctions in a pristine perovskite film, **b** comparison of the number of triple and quadruple junctions and **c** their respective proportions. Based on the statistical analysis, the quadruple junctions in a typical perovskite film only represent 3.3% of the total grain-boundary junctions.



Supplementary Fig. 9 **a** Schematic illustration of a nano-GT at the quadruple junction in a typical polycrystalline perovskite film. **b** 3D-view of the line-tension equilibrium at the quadruple junction nano-GT. **c** Detailed geometry of the nano-GT at the quadruple junction.

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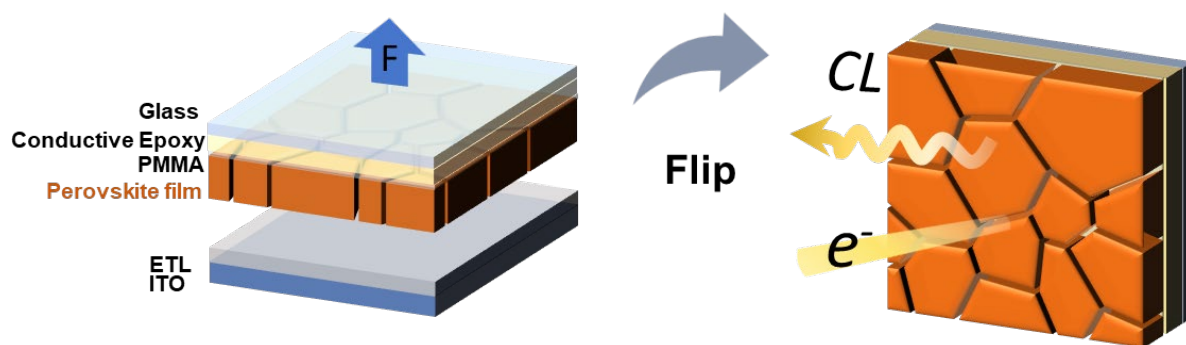
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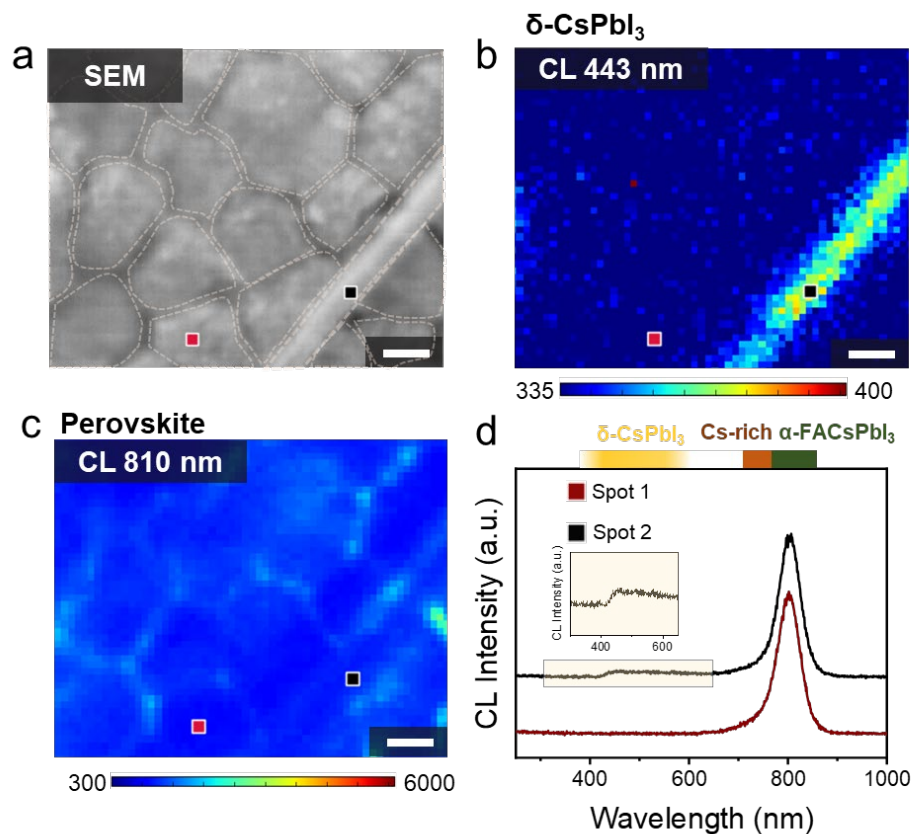
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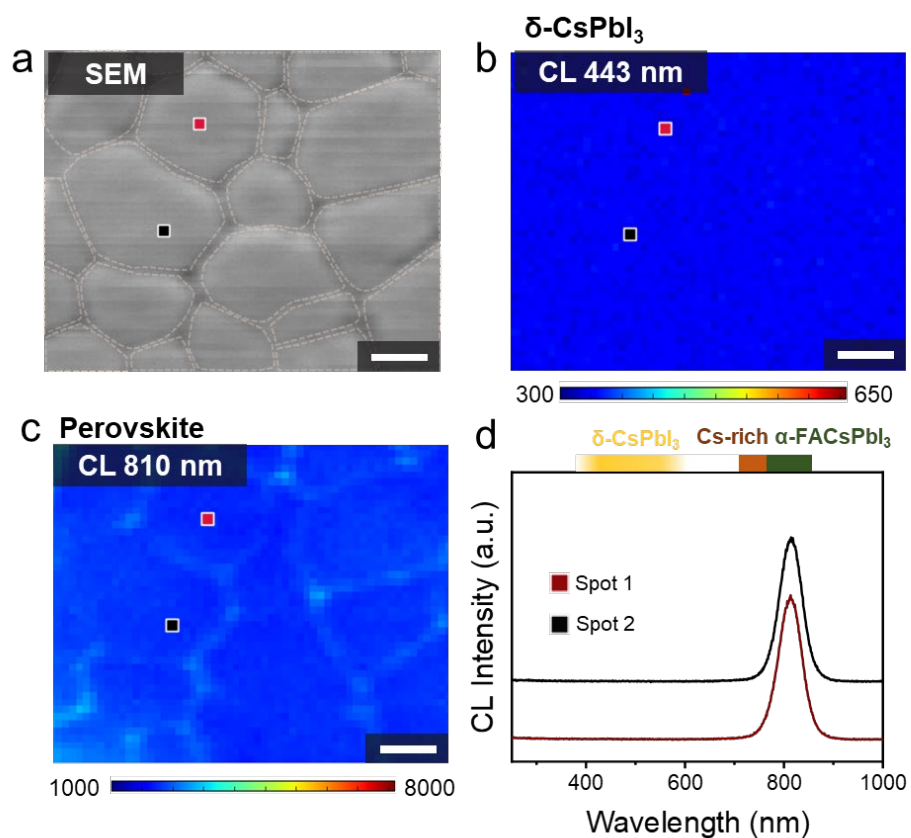
Supplementary Fig. 10 **a** Quadruple junctions in pristine perovskite films in shallowed nano-GT films and the **b** comparison of the average depth (17.7 nm for pristine and 10.1 nm for target samples) of quadruple junctions. The results show that the nano-GTs at quadruple junctions show a decrease in depth after the BAAC treatment.



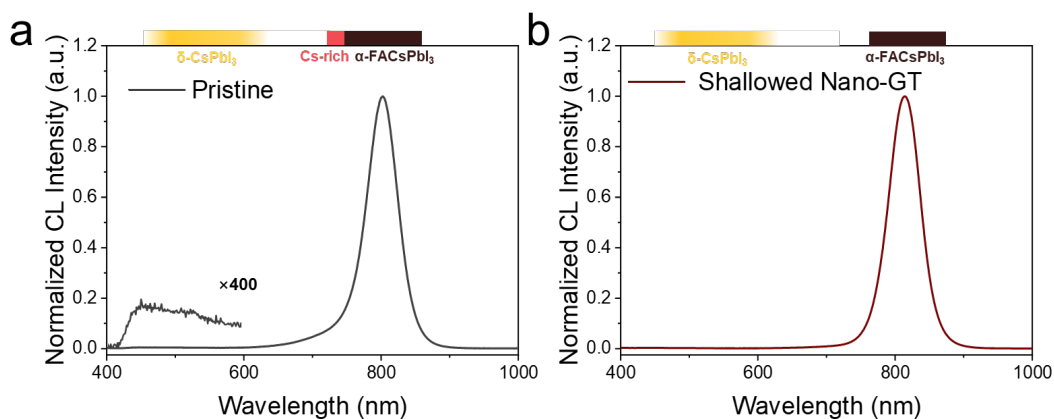
Supplementary Fig. 11 Schematic illustration of the CL measurement method based on the delaminated perovskite film sample from the ETL/ITO/Glass substrate. The sample structure is perovskite/PMMA/conductive epoxy/Glass.



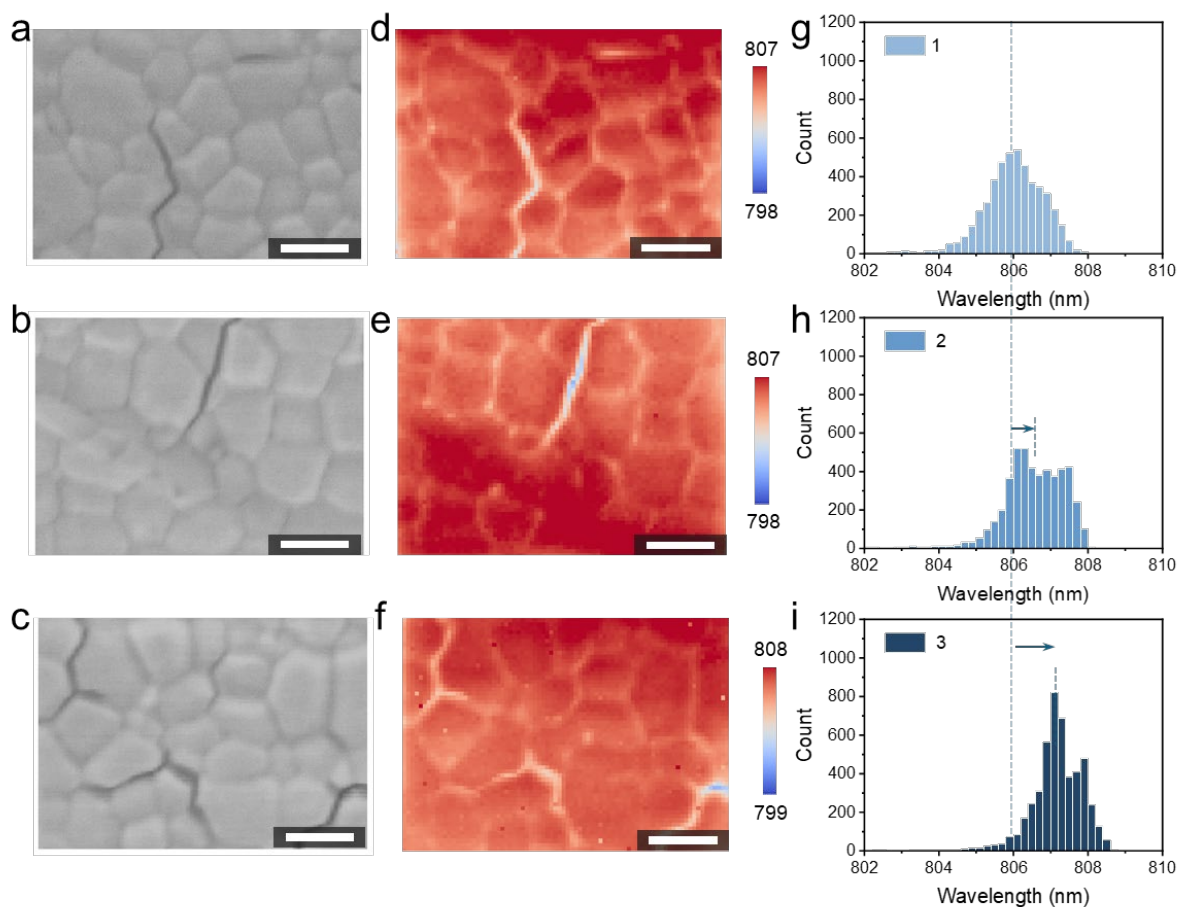
Supplementary Fig. 12 **a** SEM image, **b-c** CL microscopic maps, and **d** local CL spectra (from positions marked in the SEM image) of the pristine perovskite film. Scale bar: 200 nm.



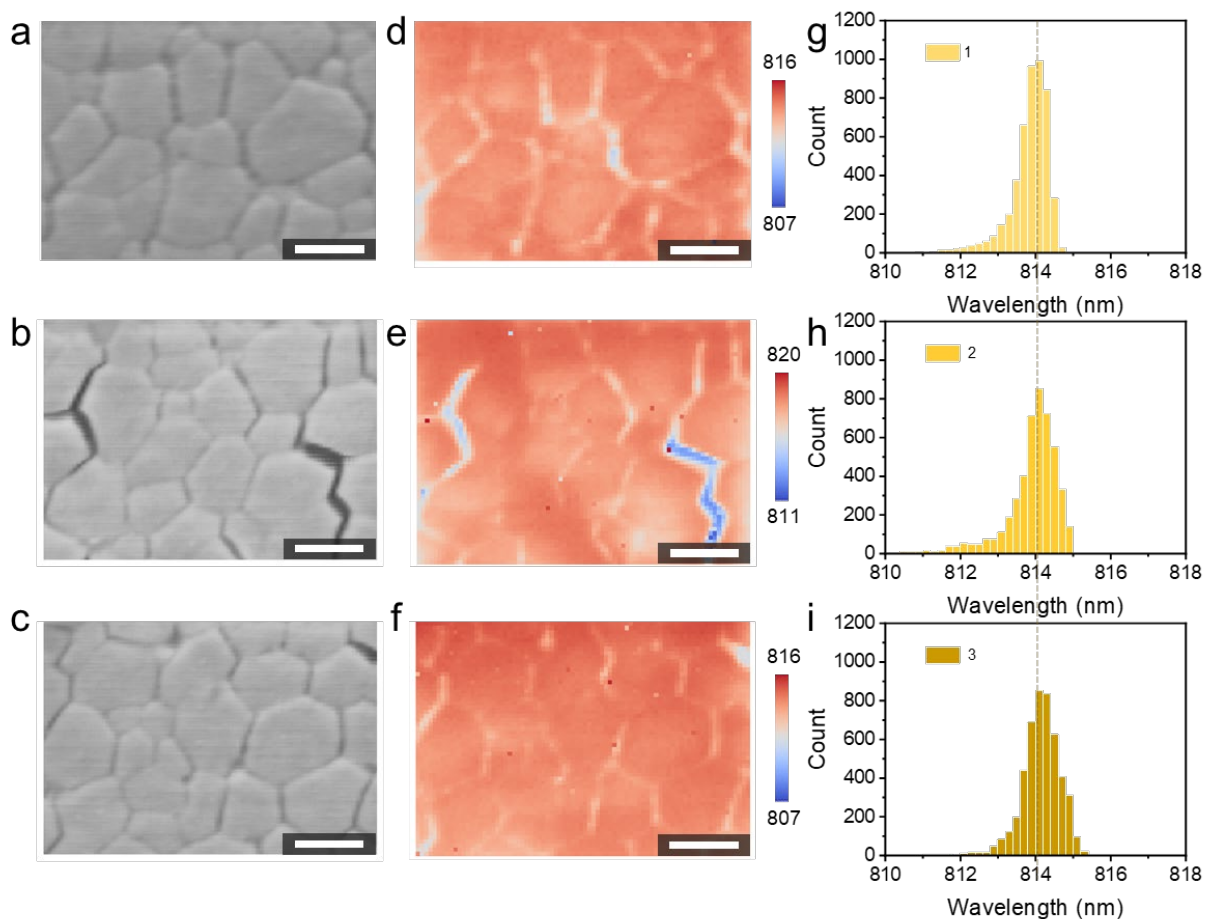
Supplementary Fig. 13 a SEM image, b-c CL microscopic map, and d local CL spectra (from positions marked in the SEM image) of the shallowed nano-GT perovskite film. Scale bar: 200 nm.



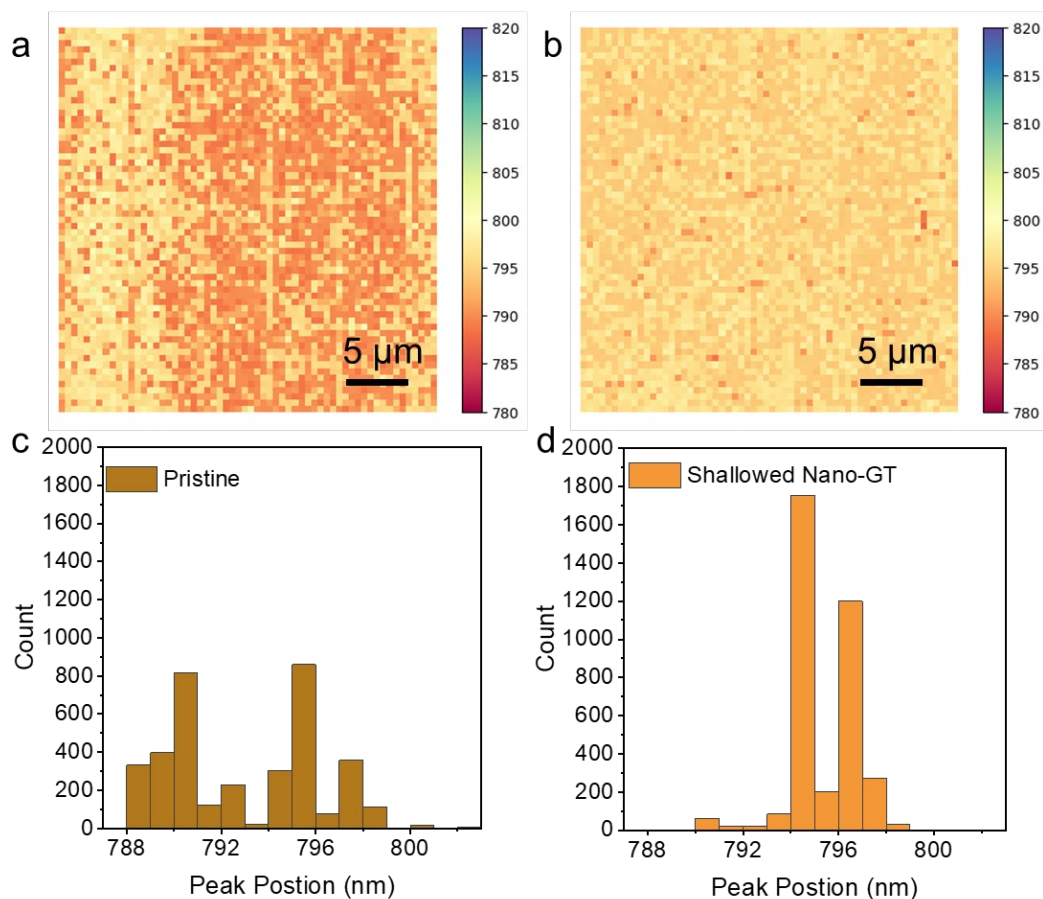
Supplementary Fig. 14 Space-average CL Spectrum of the pristine **a** and shallowed nano-GT **b** film sample. The yellow bar labels the CL emission wavelength region of δ -CsPbI₃. The red bar labels the CL emission wavelength region for Cs-rich perovskites. The black bar labels the CL emission wavelength region of the perovskite. Here we deduce that the asymmetrical peak around 810 nm in **a** is caused by the presence of Cs-rich perovskites in the film.



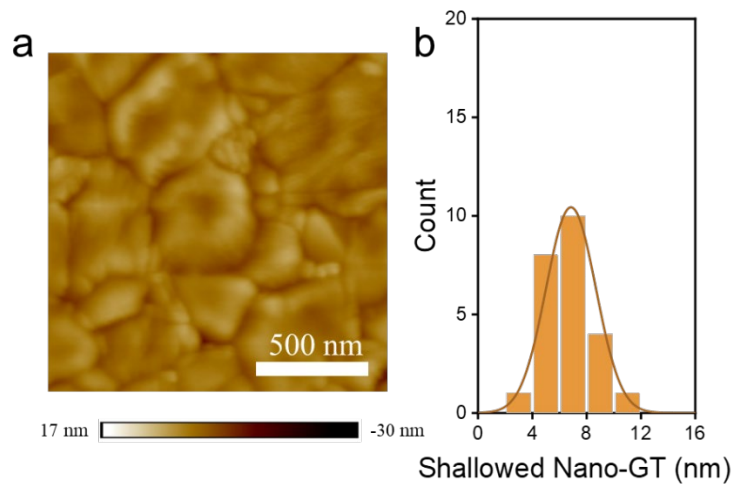
Supplementary Fig. 15 CL results obtained based on three different regions of the pristine film bottom: **a-c** SEM images, **d-f** CL mapping of the peak emission wavelengths, and **g-i** CL emission peak distributions. Scale bar: 500 nm.



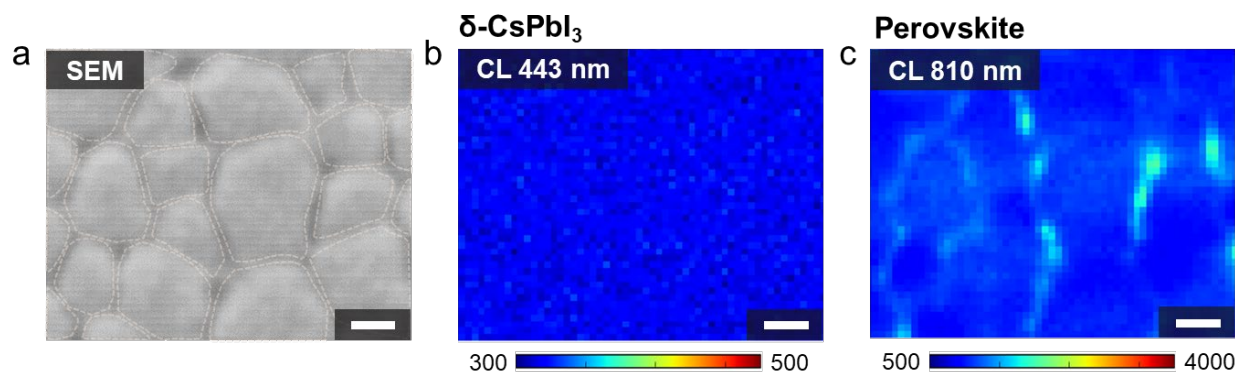
Supplementary Fig. 16 CL results obtained based on three different regions of the shallowed nano-GT film bottom: **a-c** The SEM images, **d-f** CL mapping of the peak emission wavelengths, and **g-i** CL emission peak distributions, respectively. Scale bar: 500 nm.



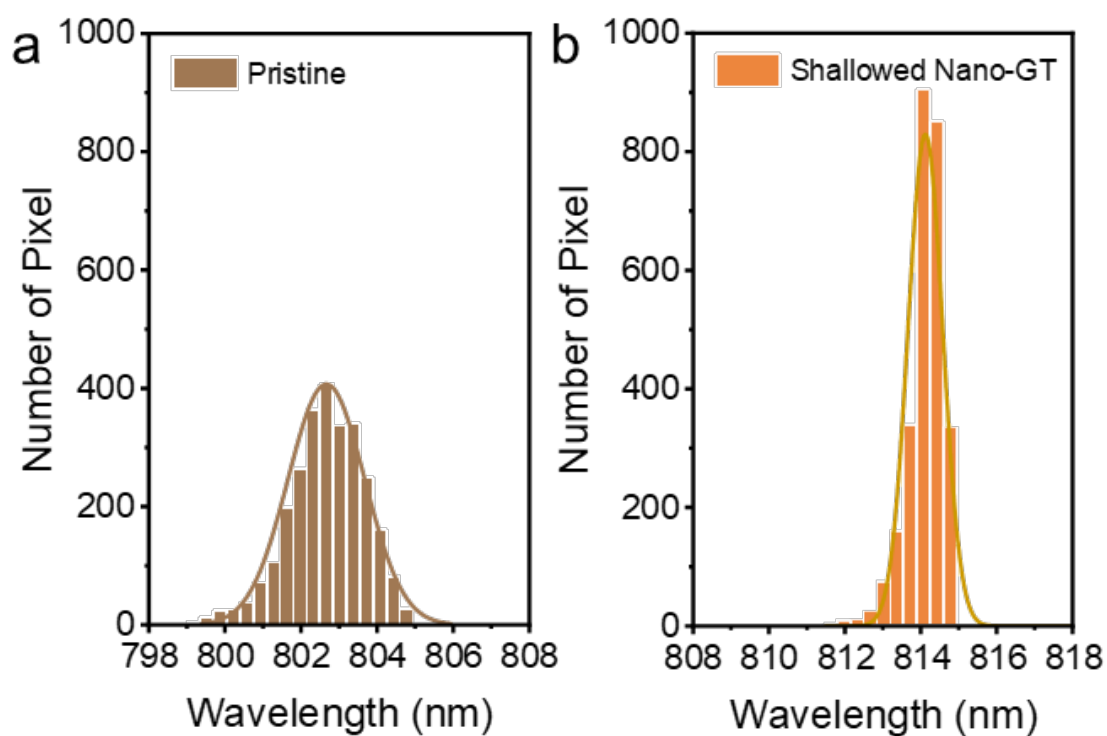
Supplementary Fig. 17 a-b Hyperspectral PL maps of the pristine and shallowed nano-GT films and **c-d** the extracted statistical distribution of PL peak positions. The hyperspectral PL pixel size of 500 nm used here is nearly equivalent to the typical grain size of our films, allowing an approximate treatment of each pixel as a grain. Spatial heterogeneity in the map of PL emission peak positions is observed in the pristine film, and the PL peak position distribution is more widely spread than that for the shallowed nano-GTs film. This further attests to the grain-to-grain cation variation in the pristine sample, which is improved in the target sample.



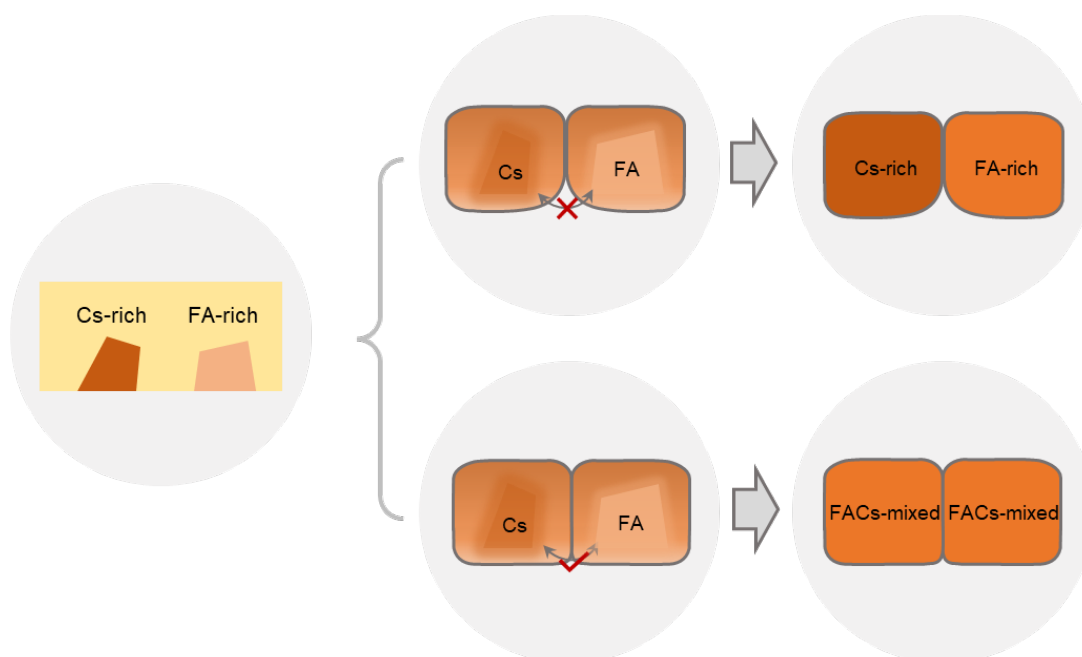
Supplementary Fig. 18 The perovskite film bottom surface with a medium-level nano-GT depth (deeper than that in the pristine film, shallower than that in the target sample): **a** AFM topographic image and **b** d_{GT} statistics based on 24 measurements.



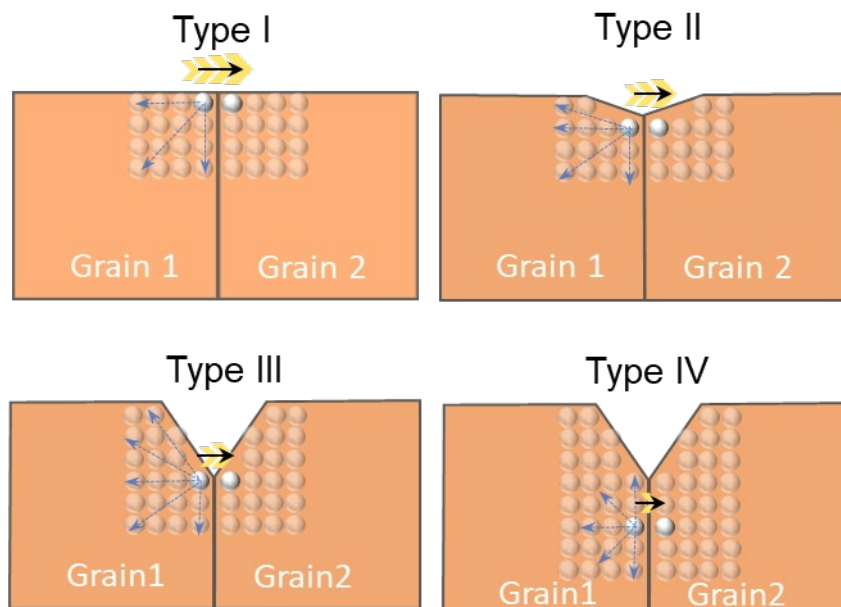
Supplementary Fig. 19 a SEM image and b-c CL microscopic maps with the 443 nm and 810 nm wavelength filters. Scale bar: 200 nm. In b, no δ -CsPbI₃ can be observed.



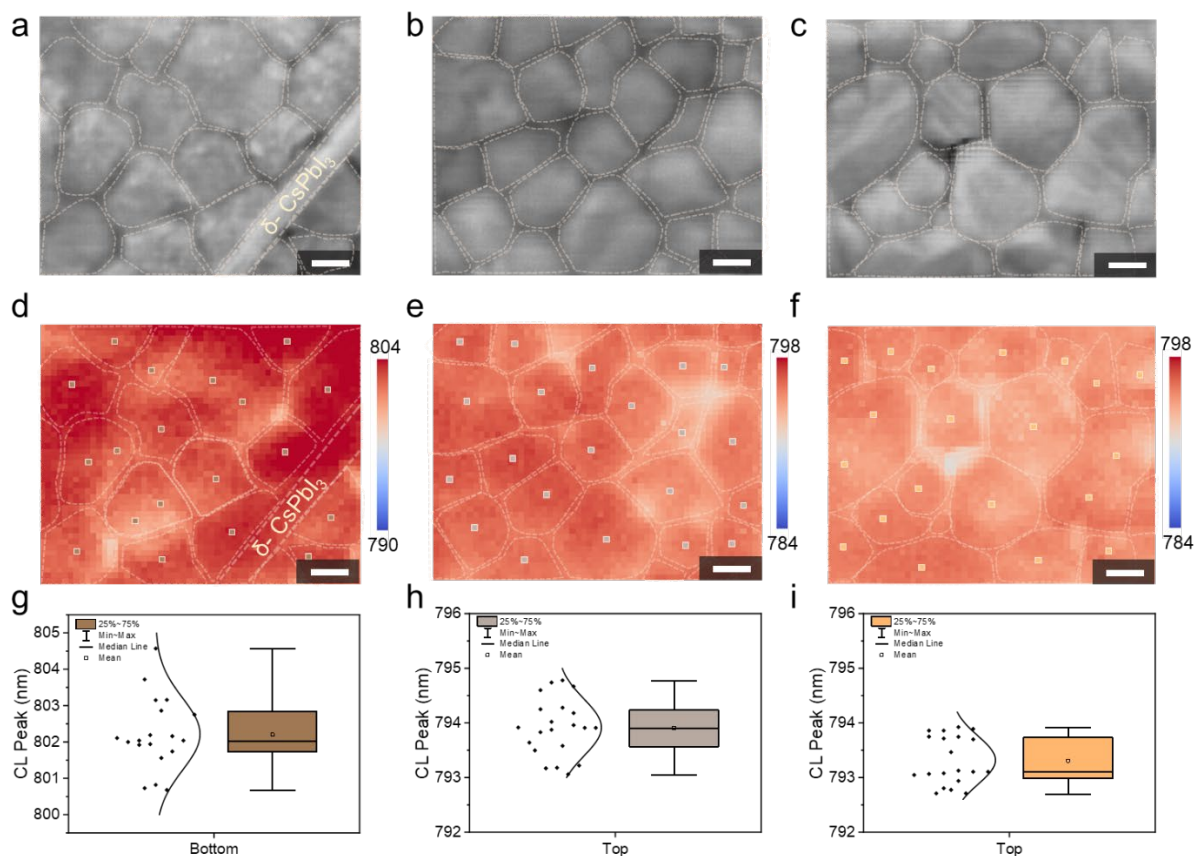
Supplementary Fig. 20 CL emission peak distributions for the pristine and shallowed nano-GT perovskite films, respectively. The distribution in the shallowed nano-GT sample is narrower than that in the pristine film. This illustrates that the CL difference in the GBs and grain interiors caused by the Cs-enriched feature has also been suppressed by nano-GT shallowing.



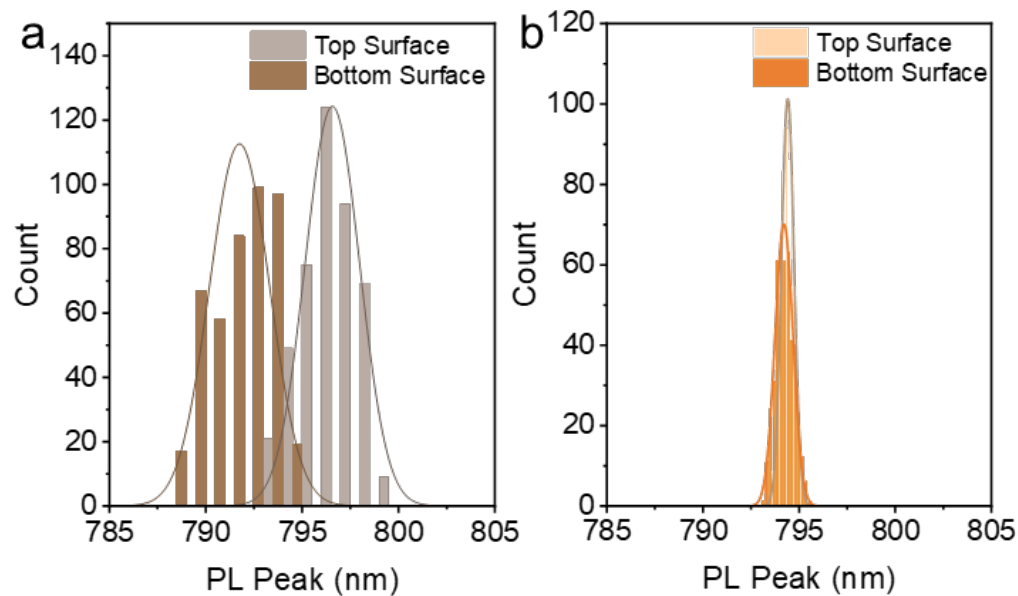
Supplementary Fig. 21 Schematics illustrating the cation interdiffusion during the FA-Cs mixed perovskite thin film growth from the solution.



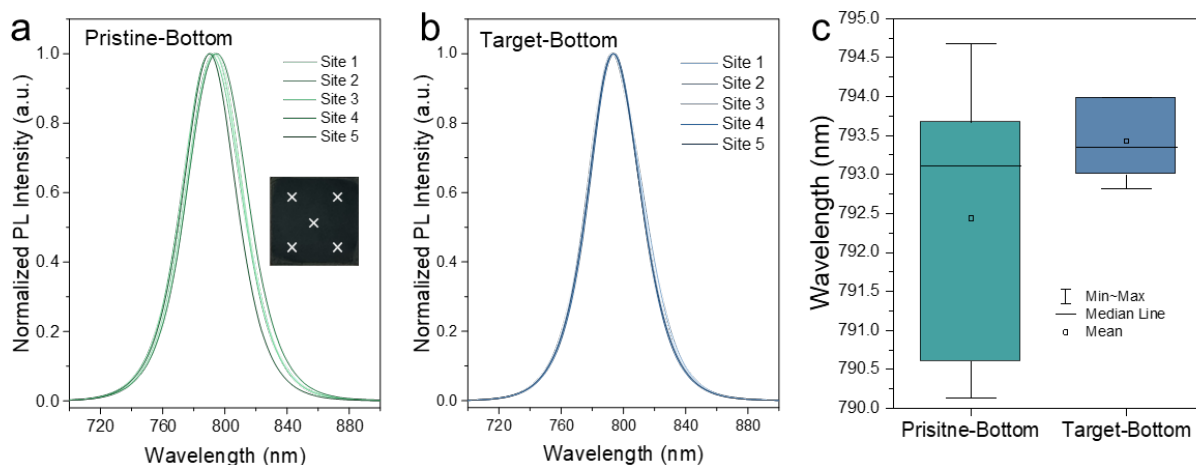
Supplementary Fig. 22 Illustration depicting atomistic scenarios of FA/Cs cation movement across various grooving conditions: (Type I) Movement across a non-grooved surface, (Type II) Movement across a less-grooved surface with reduced nano-GT depth, (Type III) Movement across a regularly grooved surface, and (Type IV) Movement across the bulk. Below, we elaborate on this illustration in detail. These 2D schematics are presented to better illustrate the atomic details in four scenarios. The quantitative physical and mathematical principles for comparing these four scenarios concerning different arrangements of neighbor atoms could be similar to analyzing the forces in a continuous version of the Schelling model.⁵ In general, the local interatomic interaction contributes to holding the moving cation back from crossing the nano-GTs. Scenario I creates the fewest neighbor atoms for the moving cation, but, unfortunately, such a non-grooving geometric configuration is thermodynamically unstable, as elaborated earlier, due to the non-equilibrium interfacial tensions. In Scenario IV, cation crossing is expected to be the most difficult, because the cation is fully constrained by the surrounding atoms in all directions. As compared to Scenario IV, the configurations in Scenarios II and III still present considerably relaxed atomistic environments for cationic motion. In this context, a shallowed nano-GT in Scenario II creates even less influence of interatomic interactions on the cation crossing, followed by intragrain diffusion. Eventually, a cross-grain cation homogenization can be achieved. Further, for a regular (deep) nano-GT, cations may even be immobilized once falling to the groove roots due to the high nano-GT angles (ζ), while in the shallowed nano-GTs, the energy barrier for de-trapping from the nano-GTs is expected to be low. The above elaborates on the behaviors of individual nano-GTs of typical types. It is expected that the nano-GTs in a film should have a statistical distribution in their characteristic depths. Even in the pristine film with nano-GT engineering, there should be relatively shallower nano-GTs that exist and allow facile cation crossing. Given sufficient time, FA-Cs cations may, in principle, diffuse via those facile paths across the whole film and eventually complete the homogenization. This is, however, impractical to perovskite because the allowed annealing time is limited before the generation of intragrain interfaces⁶ and the occurrence of thermal decomposition that can further retard FA-Cs mixing.



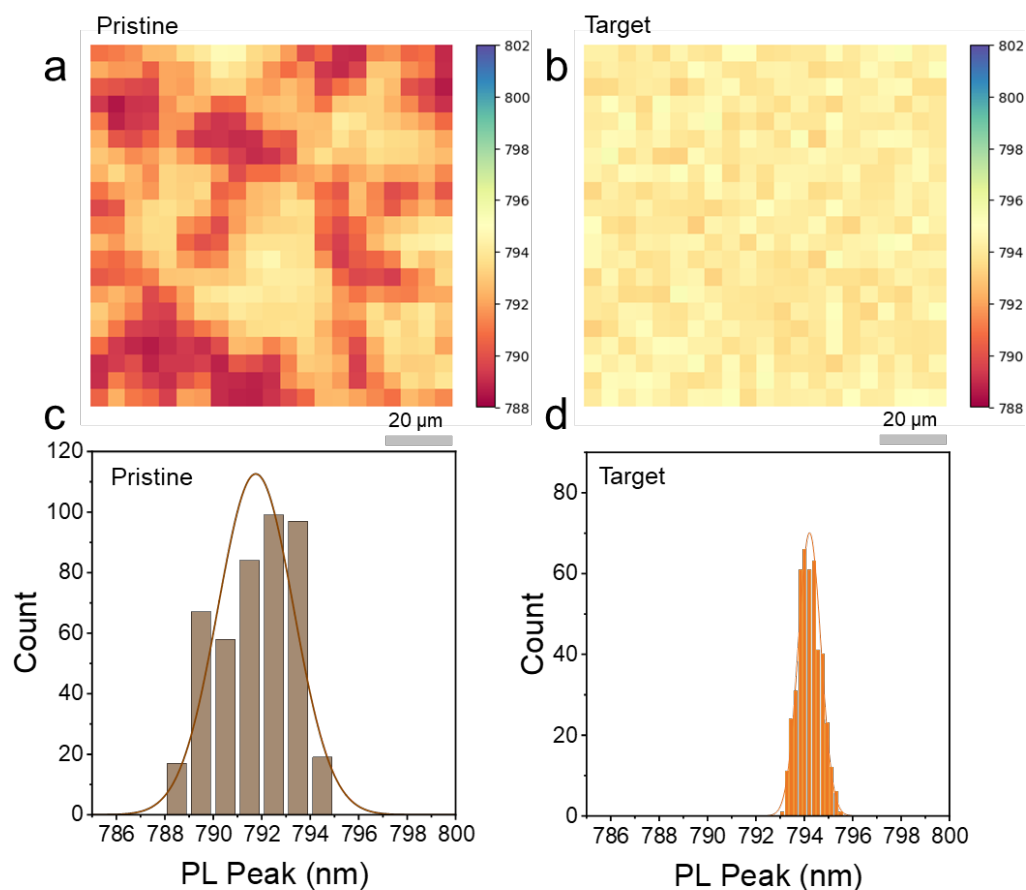
Supplementary Fig. 23 The CL results comparing the cation distributions in the bottom and top surfaces. **a-b** SEM images of the top and bottom surfaces in the pristine perovskite film, **c** SEM image of the top surface in target perovskite film. **d-f** corresponding CL emission maps (pixel size: 30 nm), and **g-i** statistical comparison of the differences in CL emissions (sample sizes of $n=19$, $n=21$ and $n=20$, respectively). CL data were collected from the marked areas in **d-f**. Each box plot displays the mean, median, upper minimum, lower maximum, and the 25% and 75% box limits, with whiskers extending to the maximum and minimum values.



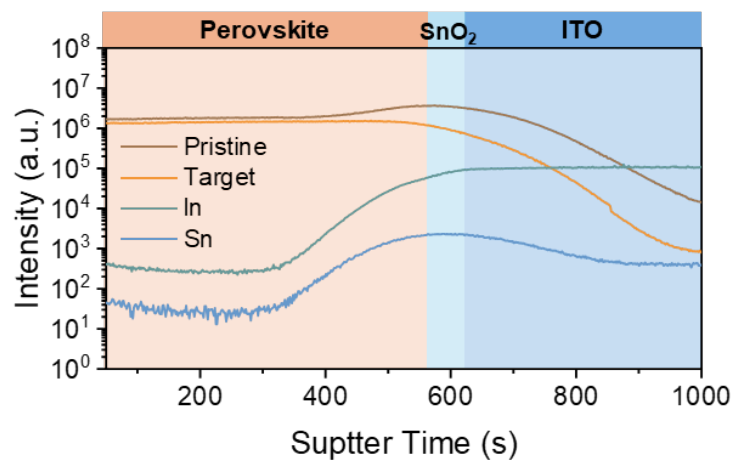
Supplementary Fig. 24 The PL emissions peak distributions measured from both top and bottom sides: **a** pristine film and **b** target film. The PL mapping covers an area of $105 \times 105 \mu\text{m}^2$, with a pixel size of $5 \mu\text{m}$ and a laser spot size of $1 \mu\text{m}$. A total of 441 measurements were taken for each film. In both pristine and target cases, the distribution of PL emission peaks on the bottom surface is relatively broader than that on the top surface. This shows, in general, that compared with the bottom surface, the top surface suffered less from grain-to-grain FA-Cs cation heterogeneity.



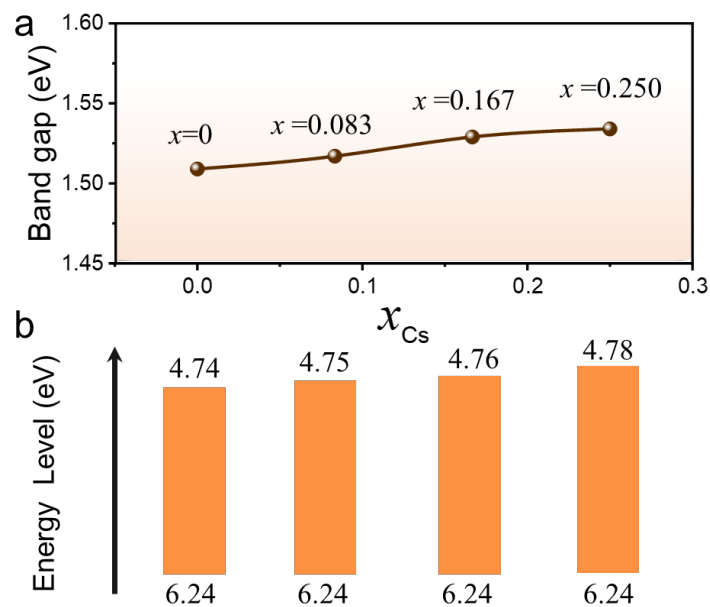
Supplementary Fig. 25 The PL results comparing the macroscopic cation homogeneity in the pristine and target films. a-b PL emissions comparisons at different locations in pristine and target bottom films ($1.5 \times 1.5 \text{ cm}^2$, inset of **a** shows the selected film locations). **c** Comparison of PL emission peaks for bottom films (sample size $n=5$ for both cases). Each box plot displays the mean, median, upper minimum, lower maximum, and the 25% and 75% box limits, with whiskers extending to the maximum and minimum values.



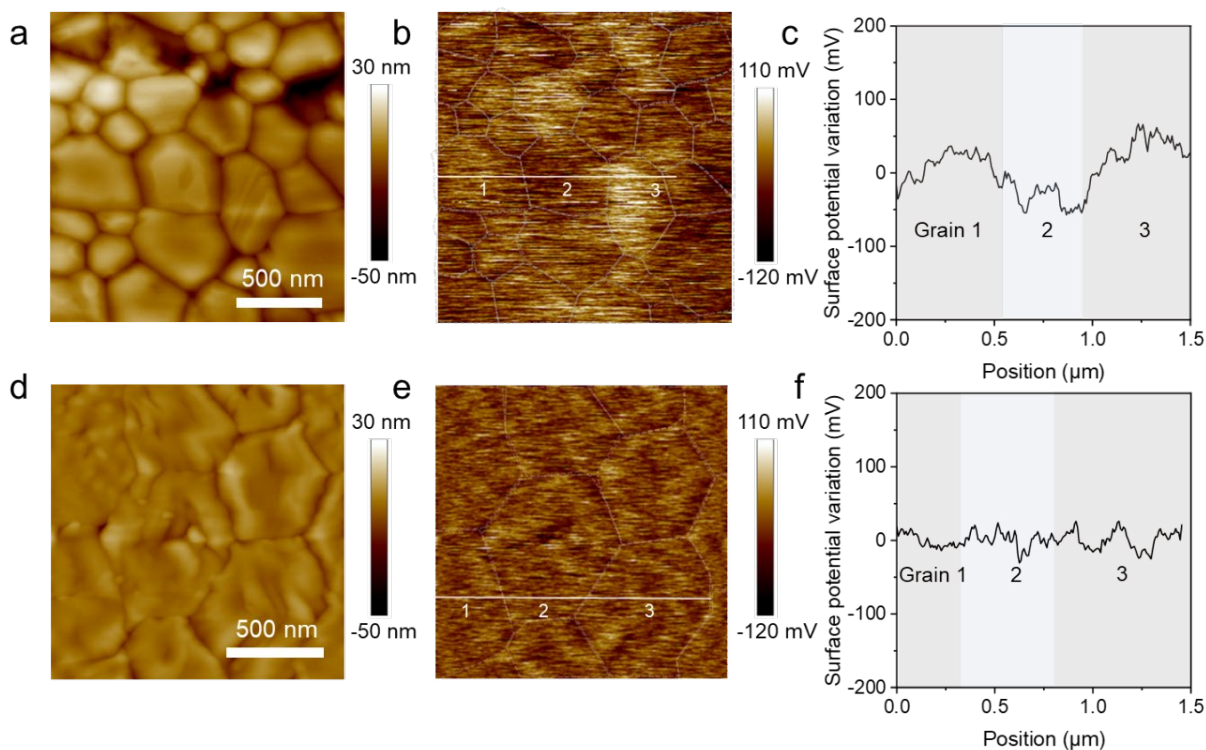
Supplementary Fig. 26 The PL results comparing the microscopic cation homogeneity in the pristine and target films. a-b The PL emission peak maps for the pristine and target film, respectively. Pixel size: 5 μm. **c-d** The corresponding PL emission peak distributions. The PL mapping covers an area of 105 × 105 μm², with a pixel size of 5 μm and a laser spot size of 1 μm. A total of 441 measurements were taken for each film.



Supplementary Fig. 27 The cross-sectional cation homogeneity in the pristine and target films. Comparison of the Cs cation distributions based on the time-of-flight secondary ion mass spectrometry (ToF-SIMS) measurements on the pristine and target films on SnO₂-coated ITO glass substrate.

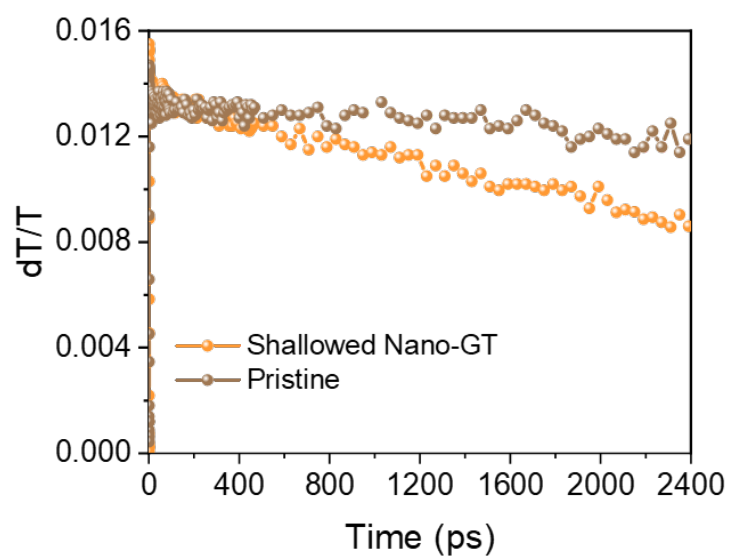


Supplementary Fig. 28 **a** Variation of the DFT calculated band gaps as a function of the composition for mixed FA-Cs in $FA_{1-x}Cs_xPbI_3$. **b** CBM (top) and VBM (bottom) alignments of corresponding compositions based on DFT calculation and Ref. [7]: from left to right, $x = 0, 0.083, 0.167$ and 0.250 .

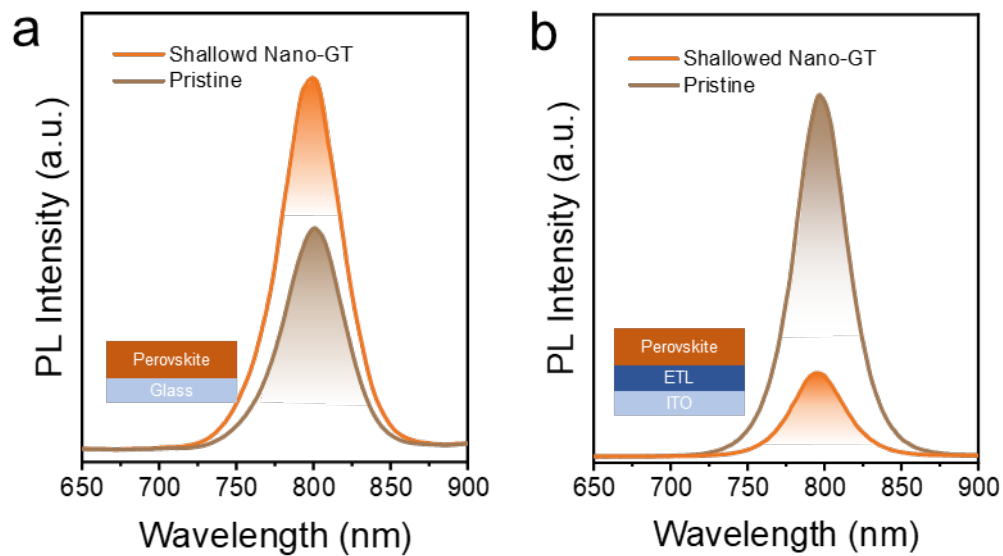


Supplementary Fig. 29 a-b The KPFM images of the pristine perovskite film. c KPFM surface potential profiles across the grains marked in b. d-e The KPFM images of the pristine perovskite film. f KPFM surface potential profiles across the grains marked in e.

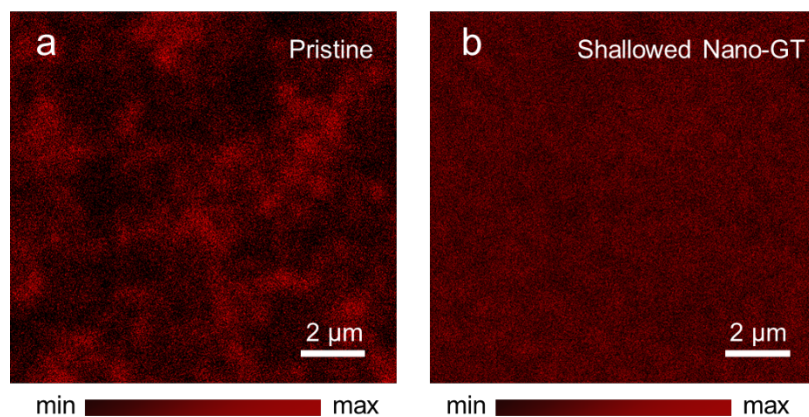
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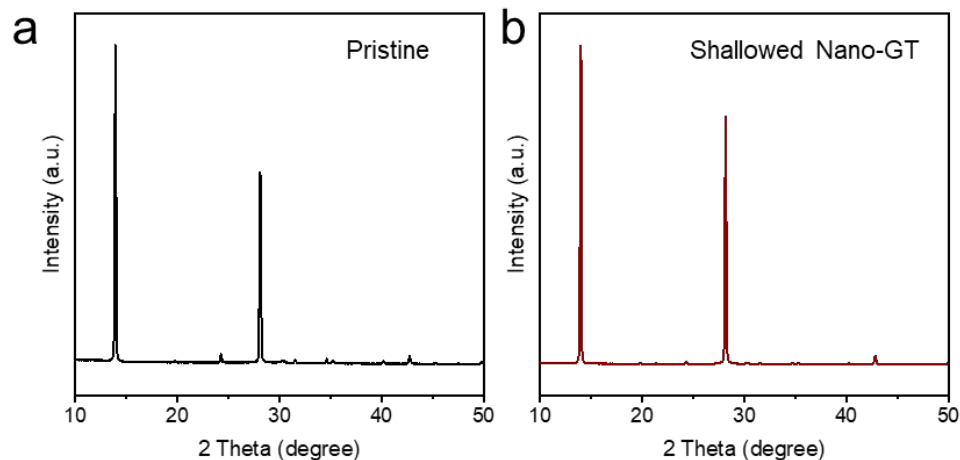
Supplementary Fig. 30 TA spectra of the pristine and shallowed nano-GT films. All the perovskite films were directly fabricated on ETL-coated substrate.



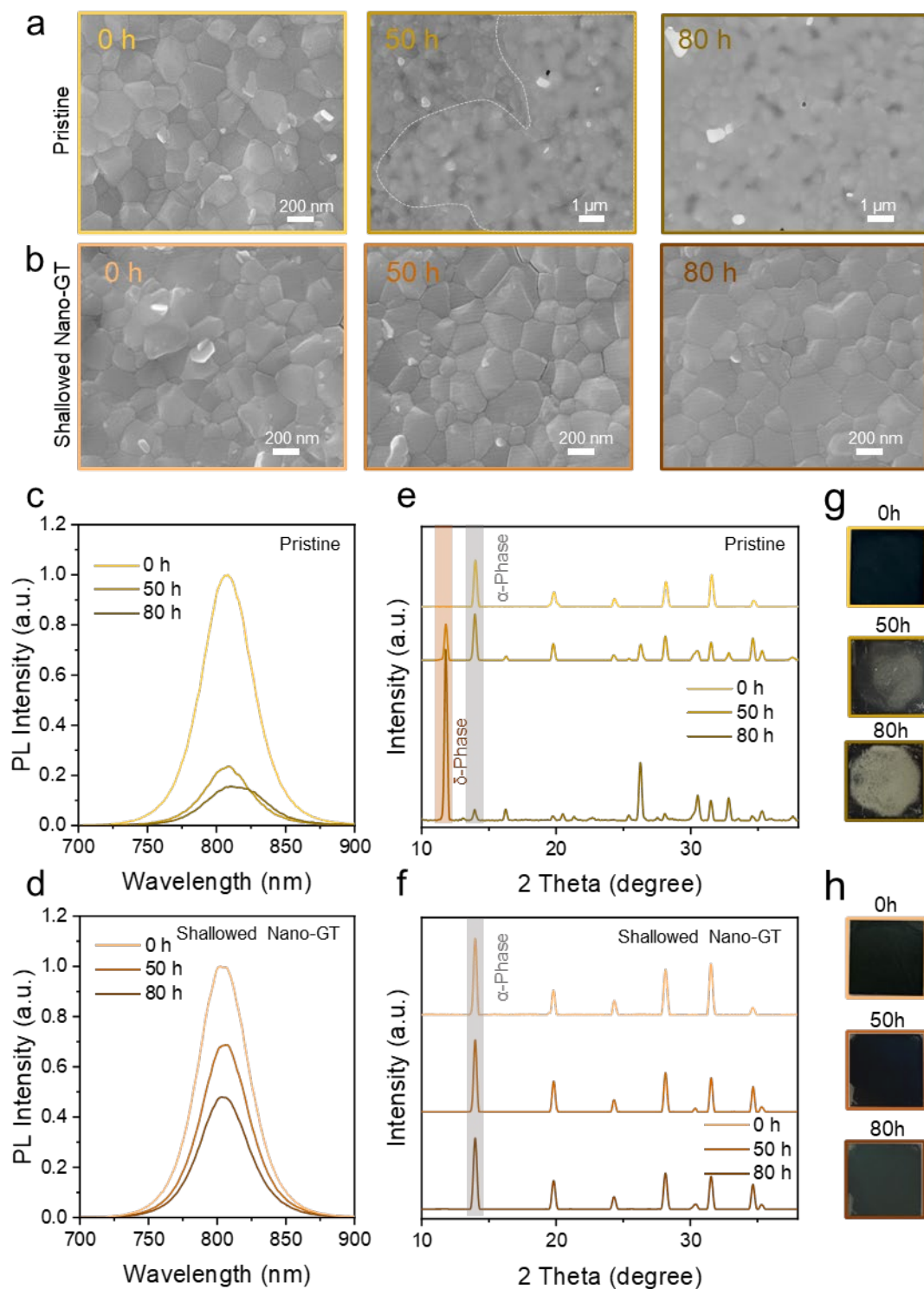
Supplementary Fig. 31 Steady-state PL spectra of the pristine and shallowd nano-GT perovskite films **a** without ETL and **b** with ETL.



Supplementary Fig. 32 a-b Large scale PL mapping of the pristine and shallowed nano-GT films on ETL-coated substrates with long pass after 726 nm wavelength filter. The intensities in the PL mappings are normalized to the peak values.

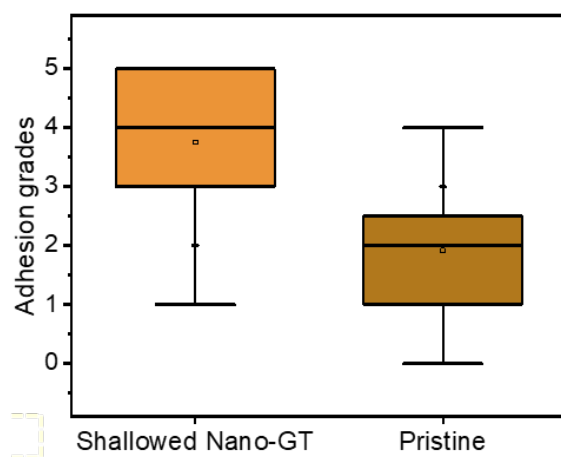


Supplementary Fig. 33 Initial XRD pattern of the pristine **a** and shallowed nano-GT **b** films before the ambient stability tests. The test conditions are 22 °C and 40 % RH. Note that the buried δ -CsPbI₃ nanophases in the pristine film bottom are in a small amount regardless of their nontrivial effects. Thus, they remain non-detectable in our standard XRD measurement primarily due to the buried nature and detection limit.



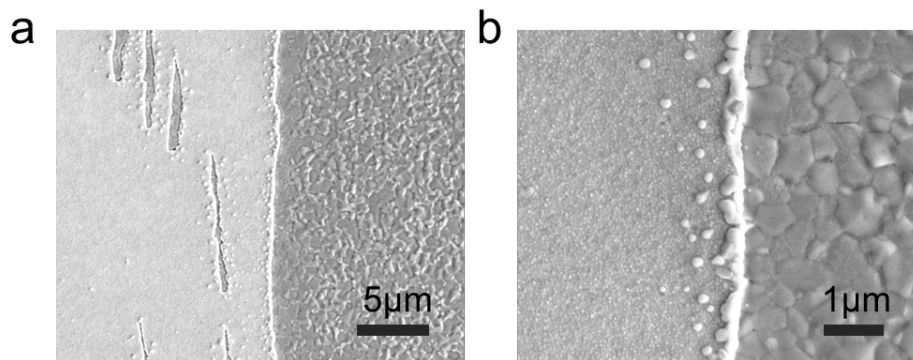
Supplementary Fig. 34 a-b SEM images, c-d PL spectra, e-f XRD patterns, and g-h optical photographs of the pristine and shallowed nano-GT sample during the stability test at 0, 50, and 80 h. The test conditions are 22°C and 80 % RH.

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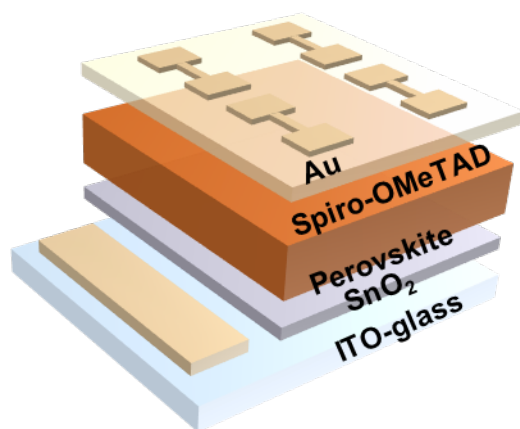


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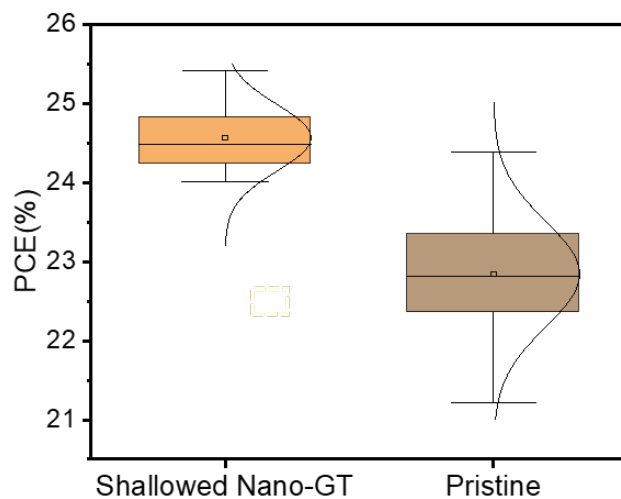
Supplementary Fig. 35 Adhesion grade statistics (sample size $n=12$ for each case) for the pristine and shallowed nano-GT films. Each box plot displays the mean, median, upper minimum, lower maximum, and the 25% and 75% box limits, with whiskers extending to the maximum and minimum values.



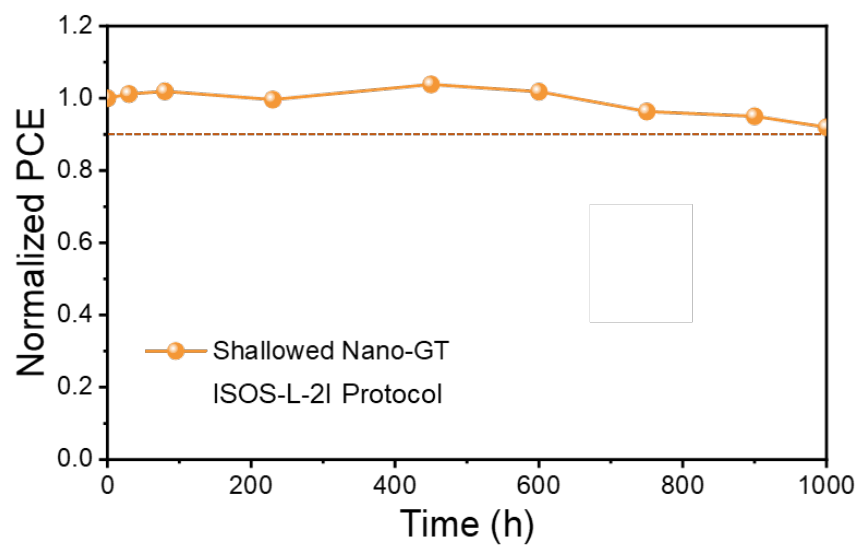
Supplementary Fig. 36 SEM images showing the existence of SnO₂ nanoparticles (delaminated from the ETL-coated substrate) on the flipped, shallowed nano-GT perovskite film.



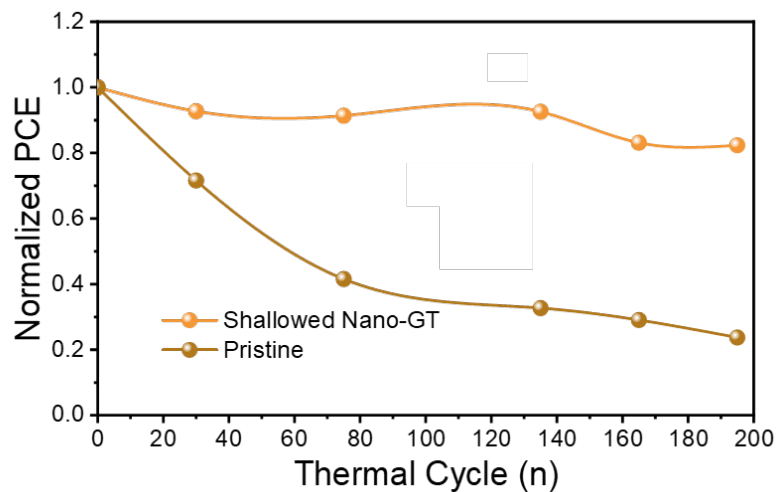
Supplementary Fig. 37 Schematics show the PSC device structure for achieving the best PCEs.



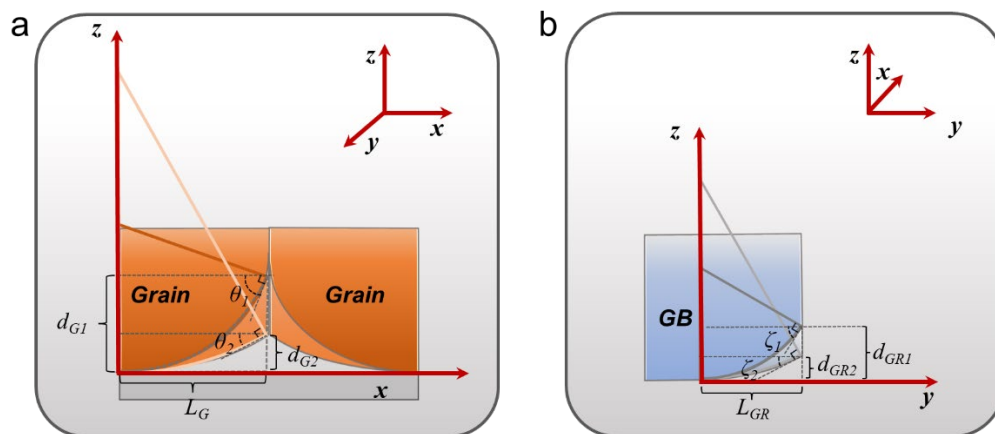
Supplementary Fig. 39 PCE statistics (sample size $n=40$ for both cases) for the pristine and shallowed nano-GT PSCs. Each box plot displays the mean, median, upper minimum, lower maximum, and the 25% and 75% box limits, with whiskers extending to the maximum and minimum values.



Supplementary Fig. 40 Light-soaking test results of PSCs with shallowed nano-GT perovskite films under the ISOS-L-2I protocol. Typically, the initial PCE was 21.27%.



Supplementary Fig. 41 Thermal-cycling (-40 to 85 $^{\circ}\text{C}$, ramp rate of ~ 80 $^{\circ}\text{C h}^{-1}$) tests for pristine and shallowed nano-GT devices with the structure of ITO/SnO₂/Perovskite/spiro-OMeTAD/Au. Typically, the initial PCEs for pristine and shallowed nano-GT PSCs were 22.13% and 23.08%, respectively.



Supplementary Fig. 43 Geometrical evolution from the **a** GB groove to the **b** nano-GT.

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Supplementary Table 1. Fitting parameters for the TRPL spectra in Fig. 2e

	A_1 [%]	τ_1 [ns]	A_2 [%]	τ_2 [ns]	τ_{avg} [ns]
Shallowed Nano-GT	12.89	14.70	87.11	204.47	202.47
Pristine	8.36	2.42	91.64	130.16	129.94

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Supplementary Table 2. Fitting parameters for the TRPL spectra in Fig. 2f

	A ₁ [%]	τ ₁ [ns]	A ₂ [%]	τ ₂ [ns]	τ _{avg} [ns]
Shallowed Nano-GT	9.04	4.38	90.60	30.42	30.03
Pristine	8.66	22.37	91.34	119.10	117.41

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Supplementary Note 1

In this note, we elaborate on the results in **Supplementary Fig. 4-7** in detail.

Herein, we first tailored the BAAC concentration from 0, to 10 mg/mL, to 30 mg/mL, which allowed us to tailor the depth of nano-GTs from 15.3 nm to 6.8 nm and 4.4 nm accordingly. However, when excessive concentrations of BAAC (50 mg/mL and 70 mg/mL) were used, there were substantial morphological reconstructions, creating unusual nanoparticle structures on the buried perovskite bottom surface (**Supplementary Fig. 4a-b**). This was later found to impact adverse effects on the device performance (**Supplementary Fig. 4c**). It is concluded that an optimal amount of BAAC leads to favorable surface reconstruction and, thus, energy modification while not introducing adverse chemical and microstructural effects. Then we further explored the effects of the additive type on the perovskite film structure. To gain more chemical insights into the effect of BAAC on the perovskite film structure, we used two other additives butylammonium iodide (BAI) and formamidinium acetate (FAAC) to clarify the separate roles of BA cation and Ac anion as possible. The AFM images of nano-GTs for the perovskite films made with BAI and FAAC, respectively (**Supplementary Fig. 5 a-b**). We found that BAI demonstrates the smallest influence on the depth of nano-GTs, while FAAC demonstrates the medium influence on the depth of the nano-GTs (**Supplementary Fig. 5 c-d**). Based on these results, we deduced that both BA cation and Ac anion can induce the surface energy change of the perovskite film bottom. BA cation interacts with perovskite surface via anchoring,^{1,2} while Ac anion interacts with perovskite surface via substitution.³ From the structure viewpoint, an Ac substitution potentially creates more compressive strain than a BA anchoring, as Ac is wholly incorporated into the crystal lattice, thus elevating the surface energy more obviously. The creation of compressive strain then becomes the origin of the interfacial energy increase that tailors the interfacial geometry. **Supplementary Fig. 6a-b** shows a shift of the characteristic X-ray diffraction (XRD) peak of perovskite to a lower angle in the target film, attesting the existence of compressive strain, and the Pb 4f shifts in X-ray photoelectron spectroscopy (XPS) indicates interaction between Ac and Pb. We envision there is space for identifying a further optimized combination of large organic cation and pseudohalide anion, which can enable us to tailor nano-GTs quantitatively. Moreover, we attempted to incorporate BAAC additive in the perovskite precursor solution instead of the ETL. We consider that BAAC in solution can influence both GBs and surfaces in resultant perovskite films, as suggested by numerous studies.⁴ We found that the BAAC addition, regardless of concentration optimization, can cause a less evident shallowing effect on the nano-GT geometry, as seen in **Supplementary Fig. 7**. When we incorporated 2 mol % BAAC into the perovskite precursor solution, the nano-GT morphologies were typical with an average depth 12.4 nm (**Supplementary Fig. 7a**), slightly smaller than the pristine one (15.3 nm). When we increased the BAAC concentration to 5 mol %, the average nano-GT depth decreased to 10.8 nm (**Supplementary Fig. 7b**). However, when we increased the BAAC concentration to 7 mol %, there was a dramatic change in the grain morphology of the perovskite bottom surface, making it hard to resolve grains and nano-GTs (**Supplementary Fig. 7c**). We overall concluded that the BAAC addition in solution causes a less evident shallowing effect on the nano-GT geometry. We expect BAAC in solution to influence both grain boundaries and surfaces in resultant perovskite films.⁴ The relative change of surface energy to grain boundary energy will be smaller, which could limit the influence of BAAC on the nano-GT geometry. This is consistent with our theoretical foundation that underpins the nano-GT geometrical engineering. Nonetheless, we do not completely rule out the possibility of further optimization and innovation of the solution-additive approach to effectively shallow nano-GTs.

Supplementary Note 2

In this note, we elaborate on the results in **Supplementary Fig. 8-10** in detail.

Herein, we would like to point out that, in addition to triple junctions, there are also quadruple junctions in the film. But quadruple-junctions only represent 3% of the total grain boundary junctions based on a statistical analysis (see **Supplementary Fig. 8**)

A similar theoretical foundation can be established for analyzing quadruple junctions. For the nano-GT at a quadruple junction, the geometry is determined by the line tension equilibrium at the interaction point of the four intersecting groove roots. **Supplementary Fig. 9a-b** presents a schematic illustration of a nano-GT at a typical quadruple junction, including the 3D view of the line tension equilibrium and the energy relationship.

$$\frac{\gamma_{QL}}{\gamma_{IS}} = 4 \sin \zeta$$

where γ_{IS} and γ_{QL} are the groove root line tension and the quadruple line tension of the nano-GTs, respectively. ζ is the angle between the groove root and the horizontal film plan. Here, ζ for the four groove roots are assumed to be identical. The energy profiles and their relations for nano-GTs are schematically illustrated in **b**. For given perovskite precursor and deposition method, γ_{QL} can be considered constant. γ_{IS} and γ_S are positively correlated as we described before. The total depth of a nano-GT (d_{GTQ}) at quadruple junctions is the sum of the grain-boundary groove depth (d_G) and the groove root depth at quadruple junctions (d_{GRQ}), as shown in **c**. Therefore, the d_{GTQ} can be described as:

$$d_{GTQ} = d_G + d_{GRQ} = \frac{l_G \cdot \cos\left(\frac{\pi}{2} - \theta\right)}{1 + \sin\left(\frac{\pi}{2} - \theta\right)} + \frac{l_{GRQ} \cdot \cos\left(\frac{\pi}{2} - \zeta\right)}{1 + \sin\left(\frac{\pi}{2} - \zeta\right)}$$

where l_G is the length of the projected length of the groove in the vertical direction and l_{GRQ} is the length of the projected length of the groove in the vertical direction. Therefore, in a nano-GT (d_{GTR}) at quadruple junctions, we found that the enhanced γ_S can increase the γ_{IS} , thereby decreasing the side angle between the GB groove (θ) and the angle between the groove root and the horizontal film plan (ζ), therefore decreased the depth of nano-GT at quadruple junctions. i.e. shallower nano-GTs. After the BAAC treatment, these nano-GTs formed at quadruple-junctions also showed a decrease in depth (**Supplementary Fig. 10**).

Supplementary Note 3

In this note, we elaborated the results in **Supplementary Fig. 23-27** in detail.

Herein, we studied the cation distribution in the top surface and across the film thickness. We found that the top surface grain-to-grain FA-Cs cation heterogeneity is relatively less significant compared to that in the film's bottom surface according to these CL results (**Supplementary Fig. 23g-h**) and PL (**Supplementary Fig. 24a**) results. We propose that this is because the formation of film top surface is not less constrained like that of film bottom surface (in close contact with substrate). For the nano-GTs formed at the film bottom, their morphologies follow the theoretical geometries due to the existence of the boundary conditions imposed by overall flat substrate. This observation of a higher level of cation inhomogeneity at the film bottom is consistent with our expected role of nano-GTs in cross-grain cation homogenization. Furthermore, we found that the grain-to-grain FA-Cs cation heterogeneity at the top surface can be alleviated if the nano-GTs are shallowed at the bottom surface (**Supplementary Fig. 23i**). This could be related to that most of the grains are across the film thickness. It is not surprising that any influences at the film bottom may tailor the film top landscape.

We further confirm that the nanoscopic cross-grain cation homogenization induced by shallowed nano-GTs is linked to macroscopic cation homogeneity in both in-plane and out-of-plane directions. For probing the in-plane FA-Cs cation homogeneities at the macro-scale, we examined both the PL emission peaks at 5 different locations on both the pristine and target films. As compared in **Supplementary Fig. 25**, the shallowed nano-GT film shows identical emission peak positions for all selected locations, while for the pristine case, notable variations of PL emission peak positions are seen. We further acquired PL emission peak maps for both the pristine and shallowed nano-GT films. As seen in **Supplementary Fig. 26**, the inhomogeneous PL emission in FA-Cs perovskite films is intrinsically present in the pristine film, which is consistent with the reported results in the literature.⁸ Once the nano-GTs are shallowed, all the PL emission maps become much more uniform. The uniformities of emission peak positions are consistent with the levels of grain-to-grain cation homogeneity in both films as revealed by the hyperspectral CL mapping.

For probing the out-of-plane FA-Cs cation homogeneities at the macro-scale, we compared PL mapping measurements on the film top and bottom surfaces of the same (prelabelled) location. As shown in **Supplementary Fig. 24b**, the PL emission peaks distributions of both sides are nearly the same for the target film, which indicates the uniformity of cations through the out-of-plane perovskite film. In contrast, the peak wavelength difference is seen in the pristine film. In the pristine film (**Supplementary Fig. 24a**), we observed there is a notable difference in PL emission positions on the top and bottom surfaces, which suggests an FA-Cs cation inhomogeneity along the out-of-plane direction. Furthermore, we performed time-of-flight secondary ion mass spectroscopy (ToF-SIMS) on both the target and control films, confirming the homogeneous out-of-plane Cs cation distribution once nano-GTs are shallowed (**Supplementary Fig. 27**).

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